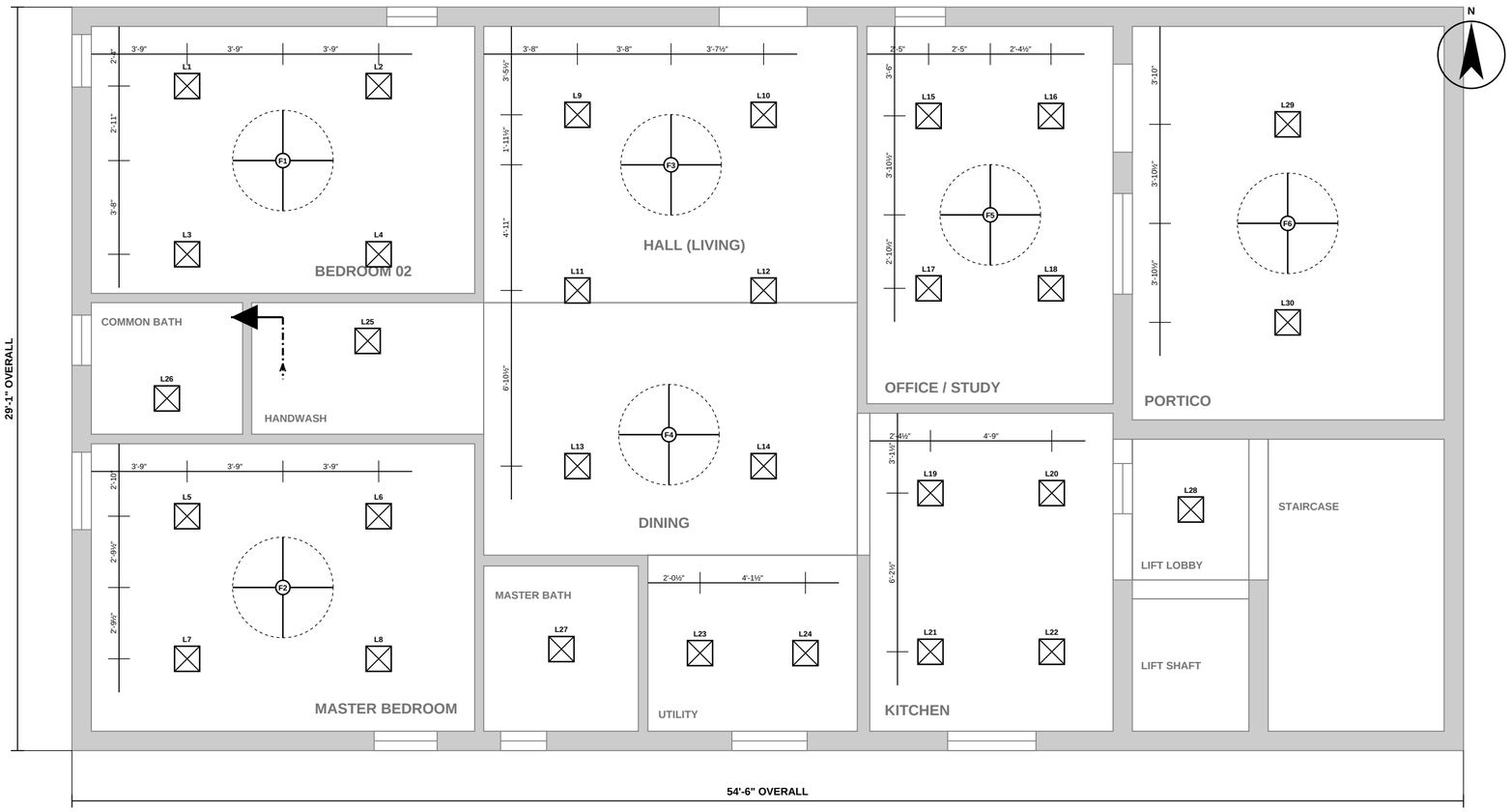
GROUND FLOOR PLAN

LIGHTING AND CEILING FAN LAYOUT • SCALE 1:50 • ALL DIMENSIONS IN FEET AND INCHES • DO NOT SCALE



SCALE 1:50

0 5 10 15 FEET

GENERAL NOTES

01 All dimensions are in feet and inches. Do not scale off this drawing.

02 This drawing is to be read with the architect's drawing ECS __/AR/R2 and the structural drawing STRUCTURAL WD-1.

03 Fixture positions are dimensioned to the centre of the fitting, from the inside faces of the north and west walls of the room concerned.

04 Clear slab soffit is 10'-6½" above ground floor finished floor level. There is no false ceiling. All luminaires and fans are surface mounted on the RCC soffit.

05 Where the first floor sunken slab lowers the soffit, the reduced level is given in the light point schedule on sheet 02. Confirm the sunken depth with the structural engineer before setting out.

06 Illumination design is to IS 3646 (Part 1) and NBC 2016 Part 8. Calculations on sheet 03.

07 Conduit: ISI marked medium gauge rigid PVC, laid between the top and bottom slab reinforcement, 20 mm dia for up to three 1.5 sq mm cores and 25 mm dia above that. Draw wire to be left in every run.

08 No conduit is to pass through any RCC beam. Route around through the slab panel or drop down the wall and rise again.

09 Fan hooks: 10 mm dia MS bar bent as a closed loop, hooked over two bottom mesh bars in each direction and tied at all four crossings, projecting 2½ in to 3 in below the soffit and capable of carrying 100 kg.

10 Junction boxes: 75 mm dia PVC, flush with the finished soffit. Maximum two 90 degree bends between draw points. No box is to be buried.

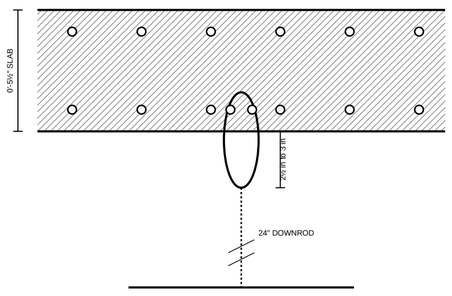
11 Contractor to confirm every fan hook and light point on site with the client before concreting, and to re-draw all fish wires within one week of the pour.

12 Task lighting, being under counter, wardrobe, mirror and desk lighting, is not part of this scope. Switching, socket outlets, distribution boards and air conditioning are covered by separate drawings.

LEGEND

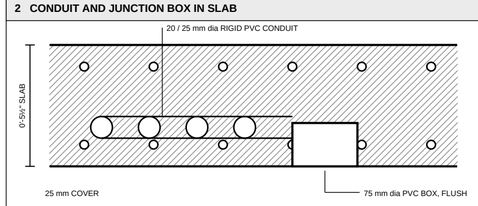
- LED SURFACE PANEL LUMINAIRE, CEILING MOUNTED — Ref Lxx. Wattage and type to luminaire schedule, sheet 02
- BLDC CEILING FAN ON HOOK CAST IN SLAB — Ref Fxx. Broken circle indicates blade sweep
- ARCHITECTURAL BACKGROUND, SCREENED — Indicative only. Refer architect's drawing ECS __/AR/R2
- SETTING-OUT DIMENSION TO FITTING CENTRE — Measured from inside face of north and west walls of that room

1 FAN HOOK CAST INTO SLAB — 1:5



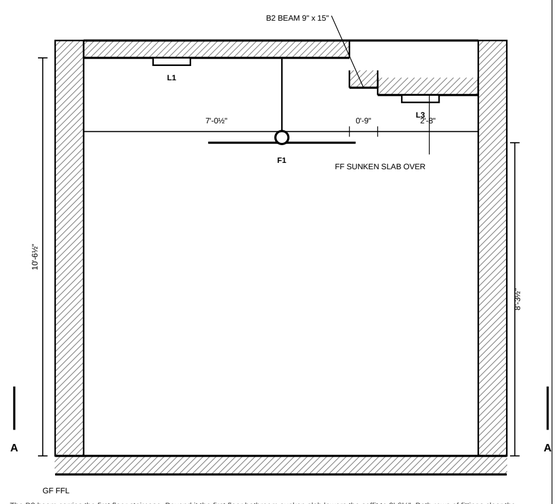
a 10 mm dia MS bar bent as a closed loop, hooked over two bottom mesh bars in each direction.
b Tie at all four crossings with 18 SWG binding wire. A hook tied to a single bar is not acceptable.
c Loop to carry 100 kg static minimum.
d Set out and fix before the top mesh is laid.

2 CONDUIT AND JUNCTION BOX IN SLAB — 1:5



a Conduit laid between the top and bottom mesh, never above the top mesh, with 25 mm minimum cover to the soffit.
b ISI marked medium gauge rigid PVC to IS 9537 (Part 3). 20 mm dia for up to three 1.5 sq mm cores, 25 mm dia above that.
c GI draw wire to be left in every run before the pour.
d Maximum two 90 degree bends between draw points. Provide an additional box rather than a third bend.
e No conduit is to pass through any RCC beam.

3 SECTION A-A • BEDROOM 02 SOFFIT LEVELS — 1:35



The B2 beam carries the first floor staircase. Beyond it the first floor bathroom sunken slab lowers the soffit to 9'-6½". Both rows of fittings clear the beam and each sits wholly on one soffit level.

Sunken depth assumed 12 in, to be confirmed by the structural engineer.

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A	22.08.2026	First issue	

SERVICES — ELECTRICAL

TO BE READ WITH ECS __/AR/R2 AND STRUCTURAL WD-1

PROJECT

ANANTAPUR, ANDHRA PRADESH

PROPOSED RESIDENTIAL BUILDING FOR MR CHANDRASEKHAR

DRAWING TITLE

GROUND FLOOR PLAN

LIGHTING & CEILING FAN LAYOUT

DRAWING NO	SCALE	DATE	REV	SHEET
GF/EL/01	1:50 @ A2	22.08.2026	A	01 of 03

DRAWN	CHECKED	APPROVED	STATUS
			FOR CONSTRUCTION

1. LUMINAIRE TYPE SCHEDULE

All luminaires surface mounted on the RCC soffit, driver integral. Minimum CRI 80, and 90 in the kitchen and bathrooms.

TYPE	DESCRIPTION	RATING	OUTPUT	CCT	IP	QTY	LOCATION
A	LED surface panel luminaire	15 W	1500 lm	3000 K	IP20	**9**	Bedroom 02, Entry / Lift Lobby, Master Bedroom
B	LED surface panel luminaire	18 W	1800 lm	3000 K	IP20	**6**	Dining, Hall (Living)
C	LED surface panel luminaire	22 W	2200 lm	4000 K	IP20	**4**	Office / Study
D	LED surface panel luminaire	18 W	1800 lm	4000 K	IP20	**4**	Kitchen
E	LED surface panel luminaire	15 W	1500 lm	4000 K	IP44	**5**	Common Bath, Handwash / Powder, Master Bath 01, Utility
F	LED surface panel luminaire	24 W	2400 lm	3000 K	IP54	**2**	Portico (covered car park)

2. CEILING FAN SCHEDULE

BLDC motor with integral RF remote receiver. Blade level held at 8'-3½" above finished floor.

REF	ROOM	SWEEP	FROM W WALL	FROM N WALL	SOFFIT	DOWNROD	BLADE LEVEL	DATUM
F1	Bedroom 02	48"	**7'-6"**	**5'-3"**	10'-6½"	24"	8'-3½"	GF FFL
F2	Master Bedroom	48"	**7'-6"**	**5'-7½"**	10'-6½"	24"	8'-3½"	GF FFL
F3	Hall (Living)	48"	**7'-4"**	**5'-5"**	10'-6½"	24"	8'-3½"	GF FFL
F4	Dining	48"	**7'-3"**	**5'-2"**	10'-6½"	24"	8'-3½"	GF FFL
F5	Office / Study	48"	**4'-10"**	**7'-4½"**	10'-6½"	24"	8'-3½"	GF FFL
F6	Portico (covered car park)	48"	**6'-1"**	**7'-8½"**	12'-6"	36"	9'-3½"	portico floor

3. CONNECTED LOAD, THIS DRAWING

ITEM	QTY	LOAD
Luminaires, general lighting	30	526 W
Ceiling fans, BLDC, 32 W each	6	192 W
Total	**36**	**718 W**

4. SUMMARY BY ROOM

ROOM	LIGHT POINTS	FANS	LOAD
Bedroom 02	4	1	60 W
Common Bath	1	—	15 W
Handwash / Powder	1	—	15 W
Master Bedroom	4	1	60 W
Hall (Living)	4	1	72 W
Dining	2	1	36 W
Office / Study	4	1	88 W
Kitchen	4	—	72 W
Utility	2	—	30 W
Master Bath 01	1	—	15 W
Entry / Lift Lobby	1	—	15 W
Portico (covered car park)	2	1	48 W

5. NOTES TO THE SCHEDULES

01 Setting out is to the centre of the fitting, measured from the inside faces of the north and west walls of the room named.

02 Soffit level is above ground floor finished floor level except in the portico, where it is above the portico floor.

03 Fittings marked at a reduced soffit level sit under the first floor sunken slab. Confirm the sunken depth before fixing.

04 IP20 fittings are for dry interiors. IP44 is required in bathrooms, the handwash and the utility. IP54 in the portico.

05 BLDC fans are controlled by their own remote. Do not fit a step type regulator on the wall plate.

06 Task lighting is not scheduled here and is by others.

6. LIGHT POINT SCHEDULE

Refer drawing GF/EL/01 for locations.

REF	ROOM	TYPE	RATING	FROM W WALL	FROM N WALL	SOFFIT	CCT	IP
L1	Bedroom 02	**A**	15 W	**3'-9"**	**2'-4"**	10'-6½"	3000 K	IP20
L2	Bedroom 02	**A**	15 W	**11'-3"**	**2'-4"**	10'-6½"	3000 K	IP20
L3	Bedroom 02	**A**	15 W	**3'-9"**	**8'-11"**	9'-6½"	3000 K	IP20
L4	Bedroom 02	**A**	15 W	**11'-3"**	**8'-11"**	9'-6½"	3000 K	IP20
L5	Master Bedroom	**A**	15 W	**3'-9"**	**2'-10"**	10'-6½"	3000 K	IP20
L6	Master Bedroom	**A**	15 W	**11'-3"**	**2'-10"**	10'-6½"	3000 K	IP20
L7	Master Bedroom	**A**	15 W	**3'-9"**	**8'-5"**	10'-6½"	3000 K	IP20
L8	Master Bedroom	**A**	15 W	**11'-3"**	**8'-5"**	10'-6½"	3000 K	IP20
L9	Hall (Living)	**B**	18 W	**3'-8"**	**3'-5½"**	10'-6½"	3000 K	IP20
L10	Hall (Living)	**B**	18 W	**10'-11½"**	**3'-5½"**	10'-6½"	3000 K	IP20
L11	Hall (Living)	**B**	18 W	**3'-8"**	**10'-4"**	10'-6½"	3000 K	IP20
L12	Hall (Living)	**B**	18 W	**10'-11½"**	**10'-4"**	10'-6½"	3000 K	IP20
L13	Dining	**B**	18 W	**3'-8"**	**17'-2½"**	10'-6½"	3000 K	IP20
L14	Dining	**B**	18 W	**10'-11½"**	**17'-2½"**	10'-6½"	3000 K	IP20
L15	Office / Study	**C**	22 W	**2'-5"**	**3'-6"**	10'-6½"	4000 K	IP20
L16	Office / Study	**C**	22 W	**7'-2½"**	**3'-6"**	10'-6½"	4000 K	IP20
L17	Office / Study	**C**	22 W	**2'-5"**	**10'-3"**	10'-6½"	4000 K	IP20
L18	Office / Study	**C**	22 W	**7'-2½"**	**10'-3"**	10'-6½"	4000 K	IP20
L19	Kitchen	**D**	18 W	**2'-4½"**	**3'-1½"**	10'-6½"	4000 K	IP20
L20	Kitchen	**D**	18 W	**7'-1½"**	**3'-1½"**	10'-6½"	4000 K	IP20
L21	Kitchen	**D**	18 W	**2'-4½"**	**9'-4"**	10'-6½"	4000 K	IP20
L22	Kitchen	**D**	18 W	**7'-1½"**	**9'-4"**	10'-6½"	4000 K	IP20
L23	Utility	**E**	15 W	**2'-0½"**	**3'-10"**	10'-6½"	4000 K	IP44
L24	Utility	**E**	15 W	**6'-2"**	**3'-10"**	10'-6½"	4000 K	IP44
L25	Handwash / Powder	**E**	15 W	**4'-6½"**	**1'-6"**	9'-6½"	4000 K	IP44
L26	Common Bath	**E**	15 W	**2'-11½"**	**3'-9"**	10'-6½"	4000 K	IP44
L27	Master Bath 01	**E**	15 W	**3'-0½"**	**3'-3"**	9'-6½"	4000 K	IP44
L28	Entry / Lift Lobby	**A**	15 W	**2'-3½"**	**2'-9"**	10'-6½"	3000 K	IP20
L29	Portico (covered car park)	**F**	24 W	**6'-1"**	**3'-10"**	12'-6"	3000 K	IP54
L30	Portico (covered car park)	**F**	24 W	**6'-1"**	**11'-7"**	12'-6"	3000 K	IP54

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SERVICES — ELECTRICAL

TO BE READ WITH ECS __/AR/R2 AND STRUCTURAL WD-1

PROJECT

ANANTAPUR, ANDHRA PRADESH

PROPOSED RESIDENTIAL BUILDING FOR MR CHANDRASEKHAR

DRAWING TITLE

GROUND FLOOR

LUMINAIRE AND FAN SCHEDULES

DRAWING NO	SCALE	DATE	REV	SHEET
GF/EL/02	**NTS @ A2**	**22.08.2026**	**A**	**02 of 03**

DRAWN	CHECKED	APPROVED	STATUS
			FOR CONSTRUCTION

1. METHOD

Lumen method to IS 3646 (Part 1) : 1992, Code of practice for interior illumination, and NBC 2016 Part 8 Section 1.

$$N = (E \times A) \div (\Phi \times UF \times MF)$$

N number of luminaires E maintained illuminance, lux A area of the working plane

Φ luminous flux per luminaire UF utilisation factor MF maintenance factor

$$K = L \cdot W \div Hm(L + W)$$

K room index L, W room dimensions

Hm mounting height above the working plane

2. DESIGN ASSUMPTIONS

PARAMETER	VALUE	BASIS
Clear slab soffit	10'-6½" above GF FFL	Floor to floor 11'-0" less 0'-5½" slab
Ceiling construction	Bare RCC, painted white	No false ceiling. All fittings surface mounted
Ceiling reflectance	0.70	White emulsion on plastered RCC
Wall reflectance	0.50	Light coloured emulsion
Floor reflectance	0.20	Vitrified tile, mid tone
Working plane	2'-5½" habitable, 2'-9½" kitchen and utility	IS 3646 (Part 1)
Maintenance factor	0.75 general, 0.70 wet and cooking, 0.65 portico	Three yearly cleaning cycle, dusty inland site
Luminaire efficacy	100 lm/W	LED surface panel, integral driver
Luminaire distribution	Near cosine, opal diffuser	UF from standard tables

3. ILLUMINATION CALCULATION

Hall and Dining are calculated as one open volume: there is no dividing wall and the beam on that line is concealed within the slab.

SPACE	SIZE	AREA	DESIGN E	IS 3646	Hm	K	UF	MF	N CALC	PROVIDED	ACHIEVED
Bedroom 02	15'-0" x 10'-5"	157	125 lx	100 min	8'-1"	0.76	0.44	0.75	3.64	**4 x 15 W**	**137 lx**
Master Bedroom	15'-0" x 11'-3"	169	125 lx	100 min	8'-1"	0.80	0.46	0.75	3.80	**4 x 15 W**	**131 lx**
Hall (Living) + Dining	14'-8" x 20'-8"	303	150 lx	150	8'-1"	1.06	0.53	0.75	5.87	**6 x 18 W**	**153 lx**
Office / Study	9'-8" x 14'-9"	142	200 lx	200 gen	8'-1"	0.72	0.43	0.75	3.74	**4 x 22 W**	**213 lx**
Kitchen	9'-6" x 12'-5"	119	200 lx	200 gen	7'-9"	0.70	0.42	0.70	4.18	**4 x 18 W**	**191 lx**
Utility	8'-2" x 6'-11"	56	150 lx	150	7'-9"	0.48	0.38	0.70	1.97	**2 x 15 W**	**152 lx**
Handwash / Powder	9'-1" x 5'-2"	47	100 lx	100	7'-9"	0.42	0.38	0.75	1.02	**1 x 15 W**	**98 lx**
Common Bath	5'-11" x 5'-2"	31	100 lx	100	10'-6½"	0.26	0.38	0.70	0.71	**1 x 15 W**	**140 lx**
Master Bath 01	6'-1" x 6'-6"	39	100 lx	100	10'-6½"	0.30	0.38	0.70	0.91	**1 x 15 W**	**109 lx**
Entry / Lift Lobby	4'-7" x 5'-6"	25	150 lx	150 lobby	10'-6½"	0.24	0.38	0.75	0.82	**1 x 15 W**	**183 lx**
Portico (covered car park)	12'-2" x 15'-5"	188	70 lx	70 covered	12'-6"	0.54	0.38	0.65	2.06	**2 x 24 W**	**67 lx**
TOTAL		**1275**								**30 nos**	**526 W**

4. LIGHTING POWER DENSITY

ITEM	VALUE	REMARK
Total lighting load	526 W	General lighting only
Total area served	1275 sft	118.5 sq m
Lighting power density	**4.44 W/sq m**	Benchmark 5 to 8 W/sq m, ECBC-R and NBC 2016

5. SPACING AND UNIFORMITY CHECK

Spacing to mounting height ratio against the maximum of 1.50 for a near cosine surface luminaire, and edge offset as a fraction of the spacing.

SPACE	SPACING X	SPACING Y	Hm	SHR X	SHR Y	MAX SHR	EDGE X / SX	EDGE Y / SY	MAX	RESULT
Bedroom 02	7'-6"	6'-7"	8'-1"	0.93	0.81	1.50	0.50	0.23	0.55	**PASS**
Master Bedroom	7'-6"	5'-7"	8'-1"	0.93	0.69	1.50	0.50	0.51	0.55	**PASS**
Hall (Living) + Dining	7'-3½"	6'-10½"	8'-1"	0.90	0.85	1.50	0.50	0.50	0.55	**PASS**
Office / Study	4'-9½"	6'-9"	8'-1"	0.59	0.84	1.50	0.50	0.52	0.55	**PASS**
Kitchen	4'-9"	6'-2½"	7'-9"	0.61	0.80	1.50	0.50	0.50	0.55	**PASS**
Utility	4'-1½"	—	7'-9"	0.53	—	1.50	0.49	—	0.55	**PASS**
Portico (covered car park)	—	7'-9"	12'-6"	—	0.62	1.50	—	0.49	0.55	**PASS**

DESIGN COMMENTARY

Effect of the high soffit

Surface mounting at 10'-6½" with no false ceiling puts the mounting height at 8'-1" above a 2'-5½" working plane in the habitable rooms. The room index falls to between 0.7 and 1.1 and the utilisation factor to between 0.38 and 0.53.

A false ceiling at 9'-0" would raise the utilisation factor by roughly one fifth and reach the same illuminance with fewer fittings. The quantities in this drawing are the direct consequence of the decision to keep the bare soffit.

Bedrooms

Both bedrooms are laid out on a symmetrical two by two grid that is deliberately not tuned to any one bed position. The bed may be placed against the south, west or north wall and the distribution remains even.

In Bedroom 02 a 9 in x 15 in beam crosses the room 7'-0½" in from the north wall, carrying the first floor staircase, and the strip beyond it lies under the first floor sunken slab. The two rows of fittings are set at 2'-4" and 8'-11" so that the beam falls in the gap between them and neither row lands on it.

Task lighting

Under counter lighting in the kitchen, wardrobe lighting, bathroom mirror lighting and desk lighting in the study are excluded from this calculation and are to be provided separately. The general levels above are therefore the ambient levels only.

Items to be confirmed

Depth of the first floor sunken slab, assumed here as 12 in. It governs the soffit level in Bedroom 02, both bathrooms and the Handwash.

Depths of the beams shown without a type mark on the structural drawing.

Staircase lighting is not included. The stairwell is open to the first floor and its fittings are carried on the first or second floor slab.

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PROPOSED RESIDENTIAL BUILDING FOR MR CHANDRASEKHAR

DRAWING TITLE

GROUND FLOOR

ILLUMINATION CALCULATIONS

DRAWING NO	SCALE	DATE	REV	SHEET
GF/EL/03	**NTS @ A2**	**22.08.2026**	**A**	**03 of 03**

DRAWN	CHECKED	APPROVED	STATUS
			FOR CONSTRUCTION